

Introduction to mathematics and optimization

January 2025

The references in the indicative answers for this exam set relate to the pdf version *Introduction to Mathematics and Optimization* (Compressed static version (5.12.2024) of interactive book).

The exam solutions must be based on the curriculum. Concepts and results in the solutions, which are not part of the curriculum, must be explained based on the curriculum.

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function

$$f(x, y) = (x + y)^2 - x - y$$

and $C \subseteq \mathbb{R}^2$ the subset

$$C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4, x + y \geq 1\}.$$

We consider the optimization problem

$$\begin{aligned} \min & f(x, y) \\ & (x, y) \in C \end{aligned} \tag{1}$$

(a) Prove that f is a convex function.

ANSWER:

The Hessian matrix for f is

$$\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}.$$

This matrix is transformed via the procedure in section 8.7 as follows:

$$\begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$$

to a diagonal matrix with non-negative entries. A diagonal matrix with non-negative entries is positive semidefinite according to Exercise 8.28. Thus, f is a convex function according to Proposition 3.41 and Theorem 8.21.

(b) Explain why C is a bounded and convex subset of $\mathbb{R}^2$.

ANSWER:

C is defined as the intersection of $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4\}$ and $\{(x, y) \in \mathbb{R}^2 \mid x + y \geq 1\}$. Thus C is a subset of $\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4\} = B((0, 0), 2)$ and C is a bounded subset according to Definition 5.29. The functions $a(x, y) = x^2 + y^2 - 4$ and $b(x, y) = 1 - x - y$ are convex. The first has the diagonal matrix $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ and the second the zero matrix as Hessian matrix (conclusion as above). Thus $C = \{(x, y) \mid a(x, y) \leq 0\} \cap \{(x, y) \mid b(x, y) \leq 0\}$ is a convex subset of $\mathbb{R}^2$ according to Lemma 4.27 and Exercise 4.23.

- (c) Show that C is compact and explain why this is important for the optimization problem (1).

ANSWER:

We know that C is bounded according to sub-question (b). From that answer, it is noted that C is closed, as a and b are continuous functions and

$$C = a^{-1}((-\infty, 0]) \cap b^{-1}((-\infty, 0]).$$

Here we have used Proposition 5.39 (the intersection of closed sets is closed), Proposition 5.42 (intervals of the form $(-\infty, a]$ are closed) and Proposition 5.51 (preimages of closed sets under continuous functions are closed). Thus C is a compact subset of $\mathbb{R}^2$ according to Definition 5.43. It is important that C is compact because the optimization problem (1) then has a solution, given that f is continuous (it is convex and thus continuous according to Theorem 6.7). This is the content of Theorem 5.66.

- (d) Explain, based on the KKT conditions, that $(2, 0)$ is not an optimal solution to (1).

ANSWER:

According to Equation (9.24), the KKT conditions for (1) are

$$\begin{aligned}\lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ x^2 + y^2 - 4 &\leq 0 \\ 1 - x - y &\leq 0 \\ \lambda_1(x^2 + y^2 - 4) &= 0 \\ \lambda_2(1 - x - y) &= 0 \\ 2x + y - 1 + \lambda_1 2y - \lambda_2 &= 0 \\ 2y + x - 1 + \lambda_1 2x - \lambda_2 &= 0\end{aligned}$$

By substituting $x = 2$ and $y = 0$ into the KKT conditions, we get

$$\begin{aligned}\lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ 0 &\leq 0 \\ -1 &\leq 0 \\ \lambda_1 \cdot 0 &= 0 \\ \lambda_2 \cdot (-1) &= 0 \\ 3 - \lambda_2 &= 0 \\ 1 + 4\lambda_1 - \lambda_2 &= 0\end{aligned}$$

We see that $\lambda_2 = 0$ and that this in the second to last equation gives $3 = 0$, which is a contradiction. Thus $(2, 0)$ cannot be an optimal solution to (1) according to Theorem 9.34(i). One notes that the optimization problem is strictly feasible, since $x^2 + y^2 - 4 < 0$ and $1 - x - y < 0$ for $(x, y) = (1, 1)$.

- (e) Show that $(1, 0)$ and $(0, 1)$ are optimal solutions to (1) based on the KKT conditions.

ANSWER:

Here Theorem 9.34(ii) and the KKT conditions are used by substituting $(1, 0)$ and $(0, 1)$. For $(1, 0)$ we get

$$\begin{aligned}\lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ -3 &\leq 0 \\ 0 &\leq 0 \\ \lambda_1 \cdot (-3) &= 0 \\ \lambda_2 \cdot 0 &= 0 \\ 1 - \lambda_2 &= 0 \\ 2\lambda_1 - \lambda_2 &= 0\end{aligned}$$

which is solved with $\lambda_1 = \frac{1}{2}$ and $\lambda_2 = 1$. For $(0, 1)$ we get

$$\begin{aligned}\lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ -3 &\leq 0 \\ 0 &\leq 0 \\ \lambda_1 \cdot (-3) &= 0 \\ \lambda_2 \cdot 0 &= 0 \\ 2\lambda_1 - \lambda_2 &= 0 \\ 1 - \lambda_2 &= 0\end{aligned}$$

which gives the same solution. Thus $(1, 0)$ and $(0, 1)$ are optimal solutions to (1) according to Theorem 9.34(ii).

(f) Is f a strictly convex function?

ANSWER:

The function f cannot be strictly convex according to Theorem 6.13, as (1) has more than one minimum. Alternatively, one can see that $f(t, -t) = 0$ for all $t \in \mathbb{R}$. Thus $f(\lambda u + (1 - \lambda)v) < \lambda f(u) + (1 - \lambda)f(v)$ does not hold for all $u, v \in \mathbb{R}^2$ and $\lambda \in (0, 1)$. One can find a counterexample by setting $u = (0, 0)$ and $v = (1, -1)$.

PROBLEM 2

Consider the system of linear equations

$$\begin{aligned} 2x + y &= 1 \\ x + 2y &= t \end{aligned} \tag{2}$$

where $t \in \mathbb{R}$.

(a) The system of equations (2) can be written in matrix form $Av = b$, where

$$v = \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 1 \\ t \end{pmatrix}.$$

Find the matrix A .

ANSWER:

The matrix is read from (2) to be

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

based on the rules of matrix multiplication.

(b) Explain why A is invertible and find A^{-1} .

ANSWER:

From Exercise 3.24 it is seen that A is invertible, since $2 \cdot 2 - 1 \cdot 1 = 3 \neq 0$. The inverse is according to the same Exercise

$$\frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}.$$

(c) Explain why (2) has a unique solution for all $t \in \mathbb{R}$ and state the unique solution.

ANSWER:

The system of equations (2) has a unique solution for all $t \in \mathbb{R}$ as A is invertible. The unique solution is

$$v = A^{-1}b = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} 1 \\ t \end{pmatrix} = \begin{pmatrix} \frac{2}{3} - \frac{t}{3} \\ -\frac{1}{3} + \frac{2t}{3} \end{pmatrix}.$$

One can also find the solution by solving the equations in (2) for x and y by substituting $y = 1 - 2x$ into the second equation and finding

$$x = \frac{2}{3} - \frac{t}{3} \quad \text{and} \quad y = -\frac{1}{3} + \frac{2t}{3}.$$

(d) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be given by

$$f(x, y, t) = (1 - 2x - y)^2 + (t - x - 2y)^2$$

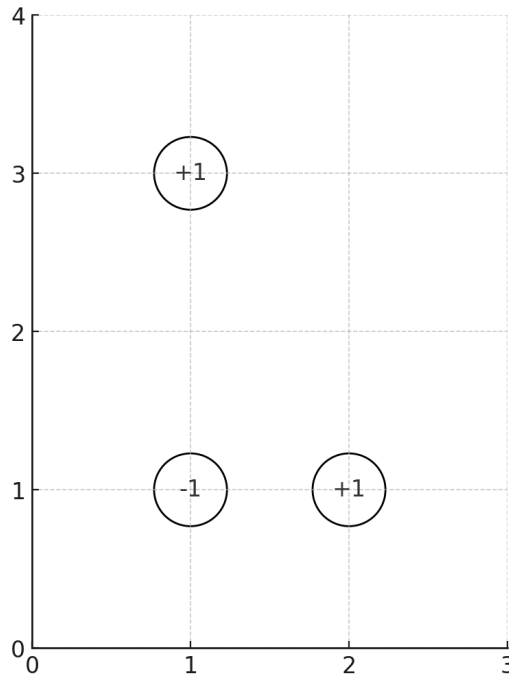
It can be used without proof that f is convex. Find the critical points for f without solving the equations $\nabla f = 0$.

ANSWER:

For a convex function, local minima are also global minima according to Theorem 6.13. Furthermore, critical points for a differentiable convex function are also minima according to Theorem 8.21 and minima are critical points according to Proposition 7.21. So we can find the critical points by finding all minima for f . But $f(x, y, t) \geq 0$ for all x, y, z, t and $f(x, y, t) = 0$ if and only if x, y, t satisfy (2). Therefore, all minima for f are precisely the points (x, y, t) that satisfy (2).

PROBLEM 3

- (a) The points $(1, 1), (1, 3), (2, 1) \in \mathbb{R}^2$ are provided with labels $-1, +1, +1$ respectively:



Show that if the line $L = \{(x, y) \in \mathbb{R}^2 \mid ax + by + c = 0\}$ separates the points, then must

$$\begin{aligned} a + b + c &< 0 \\ a + 3b + c &> 0 \\ 2a + b + c &> 0 \end{aligned} \tag{3}$$

ANSWER:

If L separates the points, it holds according to Section 9.1 (see also 5.3.2), that $ax + by + c > 0$ for all points (x, y) with label $+1$ and $ax + by + c < 0$ for all points (x, y) with label -1 . This gives the inequalities

$$\begin{aligned} a + b + c &< 0 \\ a + 3b + c &> 0 \\ 2a + b + c &> 0, \end{aligned}$$

when substituting the points.

- (b) Show using Fourier-Motzkin elimination that a solution to the inequalities in (3) must satisfy $a > 0$ and $b > 0$, and that for given $a > 0$ and $b > 0$ there exists c such that the inequalities in (3) are satisfied.

ANSWER:

We are explicitly asked to use Fourier-Motzkin elimination cf. section 4.5. We start by eliminating c :

$$\begin{aligned} c &< -a - b \\ -a - 3b &< c \\ -2a - b &< c \end{aligned}$$

This gives the inequalities

$$\begin{aligned} -a - 3b &< -a - b \\ -2a - b &< -a - b, \end{aligned} \tag{4}$$

which are summarized as $2b > 0$ and $a > 0$. Therefore, $a > 0$ and $b > 0$ hold, if the inequalities in (3) are satisfied. For given $a > 0$ and $b > 0$ we know that the inequalities in (4) are satisfied and thus that $\max(-a - 3b, -2a - b) < -a - b$. Any c between these two numbers, together with a, b , gives a solution to the inequalities in (3).

(c) Set up an optimization problem to find the best line that separates the points.

ANSWER:

The optimization problem is set up according to Theorem 9.9. Here $\alpha = (a, b)$ and $\beta = c$ and the optimization problem becomes

$$\begin{aligned} \min \quad & a^2 + b^2 \\ & a + b + c \leq -1 \\ & a + 3b + c \geq 1 \\ & 2a + b + c \geq 1 \end{aligned}$$

(d) Show that $2x + y - 4 = 0$ is the best line that separates the points.

ANSWER:

We show that $(a, b, c) = (2, 1, -4)$ is an optimal solution to the optimization problem from sub-question (c). Cf (9.24) we set up the KKT conditions for this optimization problem:

$$\begin{aligned} \lambda_1 &\geq 0 \\ \lambda_2 &\geq 0 \\ \lambda_3 &\geq 0 \\ a + b + c + 1 &\leq 0 \\ 1 - a - 3b - c &\leq 0 \\ 1 - 2a - b - c &\leq 0 \\ \lambda_1(a + b + c + 1) &= 0 \\ \lambda_2(1 - a - 3b - c) &= 0 \\ \lambda_3(1 - 2a - b - c) &= 0 \\ 2a + \lambda_1 - \lambda_2 - 2\lambda_3 &= 0 \\ 2b + \lambda_1 - 3\lambda_2 - \lambda_3 &= 0 \\ \lambda_1 - \lambda_2 - \lambda_3 &= 0 \end{aligned}$$

By substituting $a = 2, b = 1$ and $c = -4$ into the KKT conditions, we get the equations (the other conditions are satisfied):

$$\begin{aligned} \lambda_1 - \lambda_2 - 2\lambda_3 &= -4, \\ \lambda_1 - 3\lambda_2 - \lambda_3 &= -2, \\ \lambda_1 - \lambda_2 - \lambda_3 &= 0. \end{aligned}$$

These equations are satisfied for

$$\begin{aligned} \lambda_1 &= 5 \\ \lambda_2 &= 1, \\ \lambda_3 &= 4. \end{aligned}$$

Thus $(2, 1, -4)$ is an optimal solution to the optimization problem from sub-question (c) according to Theorem 9.30(ii).

(e) Is the solution to the optimization problem from sub-question (c) unique?

ANSWER:

The constraints can via Fourier-Motzkin elimination of c be reduced to

$$\begin{aligned} b &\geq 1 \\ a &\geq 2 \\ \max(1 - a - 3b, 1 - 2a - b) &\leq c \\ c &\leq -1 - a - b \end{aligned}$$

which gives $b = 1$, $a = 2$ and $c = -4$. Thus the solution is unique.